ESKWELA Database Documentation & Technical Specification

Role: Database Manager & Engineer
Project: ESKWELA – Smart Student Productivity & Academic Management System

1. Executive Summary

This document provides the complete database architecture, data dictionary, entity-relationship models, UML design specifications, and DDL SQL schema for ESKWELA. The database is designed for high performance, offline-first Firestore persistence, secure role-based access control, and scalable academic tracking.

2. Data Dictionary

Table 1: users

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Firebase Auth UID
email	VARCHAR	255	UNIQUE, NOT NULL	Login email address
full_name	VARCHAR	150	NULLABLE	Student/admin full name
student_id	VARCHAR	50	UNIQUE	Student ID card number
school	VARCHAR	200	NULLABLE	University name
course	VARCHAR	150	NULLABLE	Degree program
role	user_role	-	DEFAULT 'student'	Role: student or admin
status	user_status	-	DEFAULT 'active'	Status: active, suspended, etc.

Table 2: schedules

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Unique schedule ID
user_id	VARCHAR	128	FOREIGN KEY	Owner user ID
class_date	DATE	-	NOT NULL	Scheduled date
class_name	VARCHAR	200	NOT NULL	Course title
class_code	VARCHAR	50	NOT NULL	Course code (e.g. IT6398)
start_time	TIME	-	NOT NULL	Start time
end_time	TIME	-	NOT NULL	End time

room	VARCHAR	50	NULLABLE	Room / Lab number

Table 3: tasks

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Unique task ID
user_id	VARCHAR	128	FOREIGN KEY	Owner user ID
schedule_id	VARCHAR	128	FOREIGN KEY	Linked class schedule
title	TEXT	-	NOT NULL	Task title
due_date	TIMESTAMPTZ	-	NOT NULL	Deadline
priority	task_priority	-	DEFAULT 'MEDIUM'	HIGH, MEDIUM, LOW
completed	BOOLEAN	-	DEFAULT FALSE	Completion flag

Table 4: ai_quizzes

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Unique quiz attempt ID
user_id	VARCHAR	128	FOREIGN KEY	User ID
topic	VARCHAR	250	NOT NULL	Quiz topic / course
score	INT	-	NOT NULL	Correct answers count
total_questions	INT	-	NOT NULL	Total questions (5-15)
percentage	NUMERIC	5,2	GENERATED STORED	Score percentage

Table 5: announcements

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Unique announcement ID
admin_id	VARCHAR	128	FOREIGN KEY	Admin author ID
message	TEXT	-	NOT NULL	Broadcast message
created_at	TIMESTAMPTZ	-	DEFAULT NOW()	Timestamp posted

Table 6: focus_sessions

Column	Type	Length	Constraints	Description
id	VARCHAR	128	PRIMARY KEY	Unique session ID
user_id	VARCHAR	128	FOREIGN KEY	User ID
duration_minutes	INT	-	NOT NULL	Duration (15, 25, 45, 60)

created_at	TIMESTAMPTZ	-	DEFAULT NOW()	Completion timestamp

3. Entity-Relationship Diagram (ERD) & Cardinality

• **USERS to SCHEDULES** (1 : N): One user has zero or many scheduled classes.

• **USERS to TASKS** (1 : N): One user manages zero or many tasks.

• **USERS to AI_QUIZZES** (1 : N): One user takes zero or many AI quizzes, contributing to global leaderboard rankings.

• **USERS to FOCUS_SESSIONS** (1 : N): One user records zero or many Pomodoro focus sessions.

• **ANNOUNCEMENTS**: Standalone entity managed exclusively by admin users for global student broadcast.

4. UML & Use Case Specifications

Use Cases:
- **Student:** Register/Login, scan study load via OCR camera/gallery, view schedule (Day/Week/Month), manage tasks, take smart AI quizzes (5-15 Qs), view live leaderboard rankings, run focus timer, toggle dark theme and online auto-sync.
- **Administrator:** Authenticate admin account, access Admin Control Center, search students, manage user accounts (timed suspension, activation/deactivation, archiving), broadcast official announcements.

5. Raw SQL Schema Script (schema.sql)

```
CREATE TYPE user_role AS ENUM ('student', 'admin');
CREATE TYPE user_status AS ENUM ('active', 'pending', 'suspended', 'deactivated', 'archived');
CREATE TYPE task_priority AS ENUM ('HIGH', 'MEDIUM', 'LOW');

CREATE TABLE users (
id VARCHAR(128) PRIMARY KEY,
email VARCHAR(255) NOT NULL UNIQUE,
full_name VARCHAR(150),
student_id VARCHAR(50) UNIQUE,
school VARCHAR(200),
course VARCHAR(150),
year VARCHAR(20),
section VARCHAR(50),
role user_role NOT NULL DEFAULT 'student',
status user_status NOT NULL DEFAULT 'active',
suspension_duration VARCHAR(50),
suspended_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMPTZ
);

CREATE TABLE schedules (
id VARCHAR(128) PRIMARY KEY,
user_id VARCHAR(128) NOT NULL,
class_date DATE NOT NULL,
class_name VARCHAR(200) NOT NULL,
class_code VARCHAR(50) NOT NULL,
school VARCHAR(150),
```

```
start_time TIME NOT NULL,
end_time TIME NOT NULL,
room VARCHAR(50),
instructor VARCHAR(150),
color_type VARCHAR(30) NOT NULL DEFAULT 'BLUE',
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMPTZ,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE tasks (
id VARCHAR(128) PRIMARY KEY,
user_id VARCHAR(128) NOT NULL,
schedule_id VARCHAR(128),
title TEXT NOT NULL,
due_date TIMESTAMPTZ NOT NULL,
course_name VARCHAR(150),
priority task_priority NOT NULL DEFAULT 'MEDIUM',
completed BOOLEAN NOT NULL DEFAULT FALSE,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMPTZ,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
FOREIGN KEY (schedule_id) REFERENCES schedules(id) ON DELETE SET NULL
);

CREATE TABLE ai_quizzes (
id VARCHAR(128) PRIMARY KEY,
user_id VARCHAR(128) NOT NULL,
topic VARCHAR(250) NOT NULL,
score INT NOT NULL CHECK (score >= 0),
total_questions INT NOT NULL CHECK (total_questions > 0),
percentage NUMERIC(5,2) GENERATED ALWAYS AS (ROUND((score::NUMERIC / total_questions::NUMERIC) * 100, 2))
STORED,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMPTZ,
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE announcements (
id VARCHAR(128) PRIMARY KEY,
admin_id VARCHAR(128) NOT NULL,
school_scope VARCHAR(200),
message TEXT NOT NULL,
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMPTZ,
FOREIGN KEY (admin_id) REFERENCES users(id) ON DELETE CASCADE
);

CREATE TABLE focus_sessions (
id VARCHAR(128) PRIMARY KEY,
user_id VARCHAR(128) NOT NULL,
duration_minutes INT NOT NULL CHECK (duration_minutes > 0),
created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
```

```
FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```